

Rationale: This course introduces the basic concepts and methodologies of digital signal processing. The covered topics include signals, systems, and signal processing, frequency analysis of signals and systems, the Discrete Fourier Transform: The DFT and Digital Filters: Design of FIR and IIR filters. It also familiarizes signal analysis techniques and concepts. The motivation of the course is to develop basic knowledge of Digital Signal Processing.

Course Content	Duration (Weeks)	COs
Introduction to Digital Signal Processing, the importance and different applications of DSP in real time system, basic concepts of signals, classifications of signals, analog and discrete signals, different types of discrete-time signals, applications of discrete-time signals, different operations of signals, analog-to-digital conversion, sampling, and quantization.	3	1
Discrete time signals and systems: Discrete time signals, discrete time systems, analysis of discrete-time linear time- invariant systems. Discrete time systems described by difference equations, implementation of discrete time systems, correlation and convolution of discrete time signals.	4	2
The Discrete Fourier Transform: The DFT, Properties of the DFT, Filtering method based on the DFT, Frequency analysis of signals using the DFT	3	3
Fast Fourier Transform Algorithms: FFT algorithms, applications of the FFT algorithm. Digital Filters: Design of FIR and IIR filters. Adaptive filter	4	3

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Teaching-Learning strategy	Assessment strategy
CO1	Describe concepts of signal and its discretization process.	Cognitive/ Understand	1	2			Lecture/Slides	Mid-term
CO2	Perform correlation and convolution operations on discrete-time signals to interpret system behavior	Cognitive/ Analyze	2	2			Lecture/Slides	Assignment
CO3	Apply Fourier transform techniques to analyze and implement frequency domain signal processing and filtering	Cognitive/ Apply	3	5			Lecture/Slides	Final Exam

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Rationale: The laboratory component is designed to develop students' practical skills in implementing problem-solving approaches using signal processing techniques. Through progressive laboratory exercises, students explore various DSP methods and apply them to analyze and solve real-world engineering problems, reinforcing theoretical concepts through hands-on experience.

Course Content	Duration (Weeks)	COs
Signal interfacing and analysis using the DSP Toolbox; generation and analysis of sine, cosine, continuous-time, and discrete-time signals; implementation of basic signal operations; system modeling with impulse response and convolution; frequency-domain analysis using FFT; and noise generation and reduction using digital filtering techniques.	9	1
Study of Fourier Transform and convolution techniques applied to real-world signals (audio, communication, and sensors), enabling modeling, simulation, and analysis of practical signal processing systems.	5	2

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Teaching-Learning strategy	Assessment strategy
CO1	Apply amplitude and time modification on discrete signal.	Cognitive/ Apply	1	3			Hands-on	Lab Test
CO2	Operate frequency domain transformation and filtering on a real-world signal.	Cognitive/ Apply	3	5			Hands-on	Lab Test